What to show the operator of a free-floating arm: reach maps against rehearsal under a base tilt budget

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Abstract

An arm on a free-floating spacecraft turns the spacecraft when it moves. If the base must stay within a tilt budget, here 2 degrees, only part of the arm's reach can be used, and an operator needs to know which part. That the reach of such an arm depends on the path is known since the early 1990s, and a planning aid that previews a move and checks it before it is executed was proposed in 1992. What has not been measured is how far a map drawn for the operator can be trusted under an attitude budget, and nobody has tested people with one. We measure it in simulation for a 7 joint arm under five controllers. One move away from rest, a map pulled in to 0.8 of its length holds from all of 10 starts under the two controllers that ignore the base, and from none of 10 under the three reactionless ones, which reach further (median 0.45 m against 0.37 to 0.39 m). Over sessions of 40 moves no controller keeps it: the whole map still holds in 1 to 4 of 10 sessions, and three sessions end stuck, all under controllers with no pull back to a rest posture. A map recomputed from the current state is right and 12 to 13 s late. A map looked up from 200 stored states is instant and promises 0.05 m too much on 5% of lines. Only the rehearsal of the command itself was right every time. We describe a tool that runs the four displays and a within-subject study protocol that compares them. [study results: no participant has been run]

Keywords: free-floating manipulator, teleoperation, workspace, base attitude, operator feedback

1 Introduction

A manipulator on a free-floating base has no fixed workspace in the usual sense. Momentum is conserved, so every joint motion moves the base, and where the hand

ends up depends on the path the joints took [1, 2]. Papadopoulos and Dubowsky [3] made this exact: some points can be reached by some paths only. Reaction null-space control [6, 7] removes the base motion when the arm has joints to spare, but only for part of the reach, and the rest costs base attitude.

In practice the attitude has a budget: an antenna has to stay pointed, a camera has to keep its target. So the question an operator actually has is not "can the arm reach that" but "can the arm reach that with the base still inside its budget". The answer is a region, and the natural aid is to draw it.

We do not claim that this region moves. That follows from path dependence and is thirty years old. We also do not claim the idea of checking a move before making it: Papadopoulos [12] proposed a planner that previews the motion and asks the operator to approve it. What is missing is the size of the problem and what to do about it for the person at the controls. How wrong is a drawn map, under which controller, after how many moves, and is a map worth showing at all next to a check that is always right? Our contributions:

1. The reach of a free-floating arm under a budget on base tilt, with joint limits and collision with the bus in it, measured for a 7 joint arm under five controllers. A tilt limit during teleoperation is not new: Sato et al. [13] kept one in 1993 by slowing the arm, on a satellite with attitude control. With nothing holding the base, the limit turns into an edge of the reach, and that edge is what we measure.
2. A measured comparison of what can be shown: the map from rest, the same map pulled in, a map recomputed from the current state, a map looked up from stored states, and the rehearsal alone. One move from rest and over sessions of 12 and 40 moves.
3. A finding we did not expect: the map fails first under the controllers built to keep the base still. They reach further, park joints on their limits, and move the edge. Plain controllers reach less and keep a pulled-in map for one move, then lose it over a session.
4. A teleoperation tool and a pre-registered study design that compare the displays with people. [and the result]

2 Related work

Workspaces of free-floating arms. Umetani and Yoshida [1] give the generalized Jacobian and already separate a workspace reachable along a straight path from a given posture from one that is guaranteed. Papadopoulos and Dubowsky [3] show that singularities depend on the mass distribution and on the path taken, and define a path-dependent workspace and a path-independent one inside it, where any path works. Vafa and Dubowsky [4] give the virtual manipulator. In all of these the workspace is bounded by singularity or by link lengths. The base is free to turn. None is bounded by a budget on base attitude, and none has joint limits in it. For a planar two-link arm we checked our own code against this theory: the band of radii where dynamic singularities exist comes out at 0.41 to 2.36 m on a grid and at the same two numbers from the masses alone (Section 4.5).

A planner for the operator. The closest work to ours is Papadopoulos [12]. The operator names a target and a path. The planner checks whether the path passes through or near a dynamic singularity, previews the simulated motion, and executes after approval. If the path fails it proposes another, built from a detour through the path-independent workspace and small closed motions of the hand that turn the base on purpose. Our rehearsal is this check with a different test in it. The differences are these. The 1992 planner tests for singularity, ours for a tilt budget, joint limits and contact with the bus (collision is named there and left out). It turns the base to get somewhere, we are not allowed to. The example is a planar arm with two joints, so there is no redundancy and no reactionless control. It draws no map, so it does not ask whether a map could replace the check. No operator was tested. We do not propose alternative paths, which that paper does.

Keeping the base still. Torres and Dubowsky [5] map which joint motions disturb the base least. Reaction null-space control [6] and its flight test on ETS-VII [7] remove the disturbance where redundancy allows. Cocuzza et al. [14] minimise the reaction torque by least squares with joint limits inside the solution, tested on an air-bearing arm. These are controllers. They do not tell an operator where the controller will stop working.

A tilt limit while teleoperating. Sato et al. [13] is the nearest work on the budget itself. A ground operator drives a satellite arm, and the satellite attitude has to stay within set limits so the link to the ground holds. Their controller slows the arm as the predicted disturbance grows and feeds the disturbance forward to the attitude control. The operator works in a graphic simulator and, in their words, does so without considering the disturbance. So the limit is hidden from the operator on purpose, and the satellite has attitude control to recover with, which is why slowing down is enough. Ours has none. Slowing a free-floating arm does not change where its base ends up, so the limit cannot be hidden: some goals are simply out. Tested in a simulator, no operators compared. [read from the abstract only, get the paper]

Drift over repeated moves. A redundant arm that brings its hand back does not bring its joints back, and on a free base the attitude does not come back either. Chen et al. [15] plan motions of a free-floating arm so that hand, joints and base attitude all reset at the end of a task, and review the earlier work on repeatability of redundant arms. That work removes the drift by planning. We measure what the drift does to a displayed map over a session when the operator picks the goals and nobody plans against it. Our finding that sessions without a pull to rest posture get stuck is this known drift, seen from the operator's seat.

Showing limits to operators. Virtual fixtures [8, 9] put limits in front of users, and have user studies, but the limits are fixed in space. Coelho et al. [16] give the operator of an aerial manipulator a haptic wall at the limits of the null-space motion of the base. That wall moves with the state, the base is actuated, and the test is a numerical simulation with no users. Predictive displays show where a delayed robot will be, not where it may go.

The gap. Path dependence is known, a check before execution is known, a tilt limit in teleoperation is known, and joint drift is known. We found no work that turns the tilt limit into a reach for a base that nothing holds, none that measures how far a

map of that reach can be trusted across controllers and over a session, and none that tests people with one. This is what a dozen searches turned up, with one paper read in full and the rest from abstracts and excerpts, not a proof that nobody did it. [one systematic search before submission]

3 System

A free-floating bus carries a 7 joint arm in MuJoCo [10]. Nothing holds the bus: no thrusters, no reaction wheels. The hand is driven by resolved-rate control on the generalized Jacobian, and the spare joint is used to cancel base reaction where it can. Base tilt is counted from the attitude at the rest pose, and the budget is 2 degrees. [masses, link lengths, gains: table from the CA3 paper]

The reach is probed along 98 directions from the rest position of the hand. Along each one the hand is driven out until the tilt reaches the budget, a joint limit stops it, or the arm would hit the bus. The 98 end points form the map the operator sees, which we call the rest map. Distance along a direction is given as a fraction of the rest map, so 1 is its edge.

A command is checked by rehearsal: the move is simulated from the state the arm is really in, and refused if it ends outside the budget. This is the check of the 1992 planner [12] with the budget, the joint limits and the bus as the test. It is the only test in this paper that is always right, and it takes about 4 s for a far goal.

4 The edge moves

4.1 Measurement

We moved the hand to a random start one move away from rest, then found by bisection how far it could go along each of 20 random directions before the budget stopped it, and compared with the same search from rest. 40 starts, 800 start and direction pairs.

In 80% of pairs the edge is within 0.05 of the rest map. In 7% it is below 0.8, and the worst is 0.35 (Fig. 1, left). The bad pairs are not a corner of the space: 17 of the 40 starts and 13 of the 20 directions have at least one. They are more common on moves across the start direction (8 to 10% against 3 to 4% along it) and when the start left more tilt behind (15% above 0.8 degrees against 5% below 0.4), but neither tells you which pair will fail.

So a map drawn from rest is wrong for about one command in fourteen near its edge, and the operator cannot tell which one.

4.2 Two explanations that failed

Leftover tilt. The tilt is counted from rest, so a move that leaves tilt behind has less budget for the next one. If that were the cause, the right map would be the rest map with the budget corrected. We repeated the search with the leftover tilt forgiven. The edge did not go back: the median distance to the rest edge was 0.05 with tilt forgiven against 0.03 with it kept. Tilt is a vector and the budget is a ball around the rest

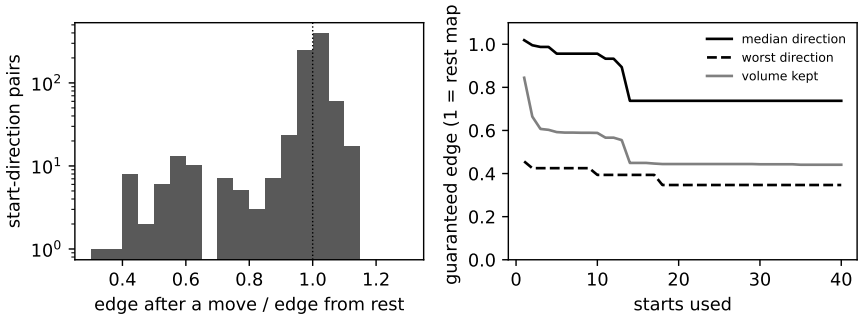


Fig. 1 Left: edge after one move, as a fraction of the edge from rest, 800 start and direction pairs. Most are at 1, a tail collapses. Right: the edge that holds from every start used so far, against the number of starts. Scripted runs.

attitude, so leftover tilt moves the edge outward in some directions and inward in others.

Enclosed area. For a free-floating system the attitude change over a closed path grows with the area the path encloses. Large shifts do only happen at large area (worst −0.29 m above 0.09 m^2, −0.05 m below 0.03 m^2), but a linear fit on signed area explains 7% of the shift.

We do not have a formula for the edge. What we have is the simulation.

4.3 Whose edge it is

Everything above is under one controller, ours: impedance on the hand with the spare joints used to cancel base reaction. We repeated it under three others: damped least squares and null-space impedance, which ignore the base, and reaction null-space control without impedance. 10 starts one move from rest, each controller against its own rest map on 98 directions.

The two reactionless controllers reach 0.24 to 0.75 m from rest, median 0.45. The two plain ones reach 0.24 to 0.64 m, median 0.37 to 0.39. Pulled in to 0.8, the map of a plain controller holds for every direction from all 10 starts. The map of a reactionless one holds for 90% of start and direction pairs and for every direction from none of the 10 starts. At 0.95 the plain maps hold for 86 to 89% of pairs, so their edge moves too, by less.

A reactionless controller arrives with less tilt spent (median 0.40 degrees against 1.07) and with joints on their limits: its failed moves end with a median of 2 joints at a limit, those of the plain controllers with none. So the far reach is bought with joint range, and what is left of the range depends on the start. This is read off counts, we have not shown the mechanism. Between 11 and 26 of the 98 lines stop short, at 0.24 to 0.30 m, with no joint at a limit, under all five controllers. Those belong to the arm, and we have not yet worked out which singularity it is.

Joint limits inside the controller. The reactionless controllers lose the map with joints sitting on their limits, so we tried one that looks at the limits while it solves, after Cocuzza et al. [14]: our controller with a weighted least-norm solve, where a joint heading for its limit costs more the closer it gets. It changed nothing. Rest

reach 0.24 to 0.75 m, median 0.46. One move out, the 0.8 map holds for 91% of pairs and from 0 of 10 starts, against 90% and 0 of 10 without it. Over 40 moves it ends at 90% of pairs and 0 of 10 sessions, with 23 of 400 moves refused. The reason is a count of joints. The hand takes three, a level base takes three, and the arm has seven, so one is left to spare the others with. In our check a joint 0.1 rad from its limit slowed by 3%. This is one velocity-level version with one untuned weight, not their acceleration-level solver, and not a controller that gives up some base tilt to save a joint. [try the version that trades tilt for joint room]

4.4 Long sessions

An operator does not make one move. We ran sessions of 40 moves, 10 per controller. Each goal is 0.2 to 0.8 of the way to the rest edge along a random direction, rehearsed, and made only if the rehearsal says yes.

After move 40 the map pulled in to 0.8 holds for 88% of directions under our controller, 83% under reaction null space, 65% under damped least squares and 85% under null-space impedance. Every direction holds in 1, 1, 3 and 4 of the 10 sessions. Null-space impedance held everything in 8 of 10 at move 20, so the loss comes late. Of the 400 goals, all well inside the map, the rehearsal refused 15, 46, 40 and 3. Two sessions under damped least squares and one under reaction null space ended with the last 10 goals all refused. Neither has a pull back to a rest posture. The two that have one did not get stuck. Tilt left behind rises over the first moves and levels off (median over the first 10 moves against the last 10: 0.31 and 0.45 degrees for ours, 0.95 and 1.22 for damped least squares), so the budget is not what runs out. The arm's posture is.

That a redundant arm drifts in posture over repeated moves is known [15]. What is new here is what it costs the map.

So no controller here keeps a fixed map over a session, at 0.8 of its length or less. The rehearsal stays right, and that is the thing an operator can trust.

4.5 A planar check

To tie the code to the known theory we built the textbook system separately: a planar arm with two links on a free base, exact momentum, no physics engine. With no tilt limit, the radii at which a dynamic singularity exists run from 0.41 to 2.36 m on a grid of joint angles. The same band from the masses and lengths alone, as the sum and difference of the three barycentric lengths, is 0.41 to 2.36 m.

5 What to show instead

5.1 A map that held from every start

Take, for each direction, the worst edge over all starts. After 40 starts the median direction keeps 0.74 of the rest edge and the worst keeps 0.35, which is 44% of the rest reach by volume. The figure was the same at 20 starts as at 40 (Fig. 1, right), so for starts one move from rest it has probably stopped shrinking. It is not a guarantee, and a second move shows it. From 40 starts two moves away from rest, one had already spent the budget on the way. From 24 of the other 39 the rehearsal refused at least one

tip of this map, 4.3% of all pairs, on 70 of the 98 directions. Pulling every direction in to 0.8 of its length leaves 11 starts and 0.9% of pairs. A fixed map cannot be made safe by shrinking it against sampled starts. Measured again on the 98 directions of the map itself, same 40 starts, the map that held keeps 0.67 of the rest edge on the median direction, 0.24 on the worst, and 40% of the volume. 28 of the 98 directions keep less than half and 5 keep everything.

5.2 A map from where the arm is

Run the 98 probes again from the current state. On one thread this takes 85 s. Shared over 10 processes it takes 12.6 s from rest and 7.4 s after a 0.3 m move, and 12 processes are no faster. So this map is right, for a hand that has stopped, and 12 to 13 s old. The tool shows a yellow marker on the hand while the map is old and a green one when it is current.

5.3 A map looked up

The map from the current state can be stored instead of computed. We took 200 states, one and two moves from rest, and for each the 98 reaches from a fast model without physics that is 8 mm off the simulation on the median. A new state is given the weighted average of its 5 nearest stored states. On states left out, the median error is 0.034 m, against 0.120 m for the average of all maps, which is what a fixed map amounts to. Against the simulated map on 5 states the median error is 0.026 m. It promises 0.05 m too much on 4.9% of lines and 0.05 m too little on 19%. It answers at once. It is not wired into the tool: one line in twenty too far is the same failure as the fixed map, less often.

Made careful. A map that promises too much is the costly kind of wrong, so we made the lookup cautious: per direction the shortest reach among the k nearest stored states, pulled in by a factor, and no reach at all when the base is already over the budget. We scored it against one fixed map given the same allowance: per direction a low percentile of the stored reaches. 1000 states, one or two moves from rest, 600 stored, 200 to pick the settings on (the furthest-reaching setting that promises 0.05 m too much on under 1% of state and direction pairs), 200 to score. On the 200 it never saw, the lookup (10 nearest, pulled in to 0.9) keeps 0.69 of the total reach and over-promises on 1.5% of pairs. The fixed map (2nd percentile) keeps 0.29 and over-promises on 1.3%. So the lookup shows more than twice the reach at about the same risk, and neither kept the 1% it was picked for. A fixed map that holds from every stored state has no reach left, because some states are already at the budget. With 200 stored states instead of 1000 the lookup kept 0.55. The reaches are the fast estimate, not the full simulation, which we have for five states only. [simulate 30 states in full and score against those]

5.4 No map

Draw nothing. Rehearse every command and refuse the ones that would break the budget. This is never wrong and gives the operator nothing to plan with: they find the edge by being refused.

6 Study

6.1 Design

Within subject, four conditions [decide whether the looked-up map is a fifth]: the rest map (the control), the map that held, the map from the current state, and no map. 20 participants, balanced Latin square over four orders, five per order. Goals come in pairs. The first is easy and moves the arm off rest, and is not scored. The second is the test and starts from where the first left the arm. Whether a goal can be reached is rehearsed from the arm's real state when it appears. A participant either puts the hand on the goal or gives it up as out of reach, at a cost of 30 s.

Measures, on the second goal of each pair: seconds per goal with the penalty and the rehearsal time in it (primary), commands the fence had to bring back, reachable goals given up, unreachable goals given up, worst tilt, and after each block raw NASA-TLX [11] and one trust rating from 1 to 7.

Hypotheses, written before data: the rest map has the most commands brought back (H1); the map that held has none and the most reachable goals given up (H2); the current-state map is the fastest per goal (H3); no map has no commands brought back but is slower and rated as more work (H4). Friedman test over conditions on the primary measure, Wilcoxon pairs with Holm correction.

6.2 Results

[No participant has been run. Ethics approval and consent to settle first. A scripted operator will be run through all blocks before people are, to see whether seven scored goals per block can separate the conditions at all; that run will be reported as scripted, not as data.]

7 Limits

Simulation only. One arm, one bus mass, one budget. The controller comparison and the sessions are 10 starts or 10 sessions per controller. No recovery move was tried on a stuck session. The map that held is tested on one-move starts. The delay of the current-state map is the delay on one laptop. The reach is probed on 98 lines from one point, so the map is a set of lines and not a volume with a proof behind it. No contact, no flexible parts, no sensor noise.

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